Precept 8 Outline

ECO 202 TAs

November 12, 2026

1 Weekly Check In

1. Any questions about the class, comments?

2 Precept practice problems

1. Suppose independent customer purchase amounts have population mean 42 dollars and population standard deviation 18 dollars. The population distribution is right-skewed. Let $\bar{X}_n$ be the mean purchase amount in an independent sample of n customers.

   (a) Calculate $E[\bar{X}_{64}]$ and $SD(\bar{X}_{64})$ exactly under the stated model.

   (b) Use the CLT to approximate $P(\bar{X}_{64} > 45)$. Show the standardized boundary and interpret the probability in repeated-sampling terms.

   (c) Repeat the calculation for n = 256 and explain why the upper-tail probability shrinks even though the population mean and threshold remain unchanged.

   (d) State one simulation that could assess finite-sample accuracy for both sample sizes if the complete population model were available. Explain why right skew makes the check useful and what the simulation would not establish about real customers.

2. A 100-order sample has mean 12.60 minutes and sample standard deviation 2.50 minutes. Use the large-sample estimated standard error $2.50/\sqrt{100} = 0.25$ minutes and the approximation $Z \sim N(0,1)$ under $H_0 : \mu = 12$. You may use $P(N(0,1) \geq 2.40) = 0.0082$.

   (a) Calculate the standardized statistic and interpret its sign and magnitude.

   (b) Sketch or describe the two reference-distribution tails that count under $H_a : \mu \neq 12$.

   (c) Calculate the two-sided p-value.

   (d) Interpret that p-value as a probability under the null and procedure; do not assign a probability to the null hypothesis.

   (e) Apply the two-sided rule at $\alpha = 0.05$ and $\alpha = 0.01$. Use reject or fail-to-reject language.

3. A randomized outreach study estimates an average intention-to-treat effect δ on an application-completion rate, measured in percentage points. For this planning exercise, treat the estimator's reference model as

$$\hat{\delta} \sim N(\delta, 2^2)$$

   Test $H_0 : \delta = 0$ against $H_a : \delta \neq 0$ at $\alpha = 0.05$ using $Z = \hat{\delta}/2$ and the rule $|Z| > 1.96$. The reference standard error 2 is treated as fixed across the alternatives considered.

   (a) Describe a Type I error in the outreach setting.

   (b) Describe a Type II error at the specific alternative $\delta = 3$ percentage points.

(c) Translate the rejection rule into critical values for $\hat{\delta}$ in percentage points.

(d) Under $\delta = 3$, show that $Z \sim N(1.5, 1)$ and calculate power using $P(N(0,1) > 0.46) = 0.3228$ and $P(N(0,1) < -3.46) = 0.0003$.

(e) Calculate the Type II error probability at $\delta = 3$.

(f) Suppose four times as many units under an otherwise comparable independent design halves the reference standard error to 1. Under $\delta = 3$, use $P(N(0,1) > -1.04) = 0.8508$ and a negligible opposite tail to approximate power. Explain the change.

(g) With design and alternative fixed, explain how lowering α affects Type I error and power.

3 Extra practice problems

1. Suppose (Y, X) has the joint distribution

$Y \backslash X$	0	2
-1	0.00	0.20
1	0.50	0.30

and let (X_i, Y_i), $i = 1, 2, \ldots$, be independent pairs, each distributed like (X, Y).

(a) Find $\mathbb{E}[X]$ and $\mathrm{Var}(X)$.

(b) Use the CLT to approximate

$$\mathbb{P}\left(\sum_{i=1}^{100} X_i > 111\right).$$

2. Suppose X has distribution

x	1	2	3
$\mathbb{P}(X = x)$	0.3	0.2	0.5

and $X_1, \ldots, X_{100}$ are independent and distributed like X.

(a) Find $\mu = \mathbb{E}[X]$ and $\sigma^2 = \mathrm{Var}(X)$.

(b) Use the CLT to approximate

$$\mathbb{P}\left(\frac{1}{100}\sum_{i=1}^{100} X_i > 2.3\right).$$

3. A study tests the null hypothesis that 70% of all U.S. companies grant their executives stock options. In a simple random sample of 420 companies, 281 grant options. Using the large-sample estimated standard error, find the p-value of an appropriate two-sided hypothesis test. State the hypotheses and your decision at the 5% level.

4. Let p_1 and p_2 be the fractions of individuals with a characteristic in two populations. Two reasonably large independent samples yield an estimated difference $\hat{p}_2 - \hat{p}_1 = 0.15$ with estimated standard error 0.10. Find the approximate p-value of the two-sided test of $H_0 : p_1 = p_2$, and state the decision at the 5% level.

5. We observe two independent random variables $X \sim N(\mu, 2)$ and $Y \sim N(\mu, 5)$, where the second argument is the standard deviation. We test $H_0 : \mu = 0$ against $H_a : \mu > 0$. Define

$$\hat{\mu}_1 = \frac{X + Y}{2}, \qquad \hat{\mu}_2 = 0.6X + 0.4Y.$$

(a) Find the critical value for the 5% level test that rejects for large values of $\hat{\mu}_1$.

(b) Consider instead the test that rejects when $\hat{\mu}_2 > 1.2$. Find its power when $\mu = 1$.

6. We observe independent random variables

$$X \sim N(\mu, 1), \qquad Y \sim N(2\mu + 1, 3),$$

where the second argument is the standard deviation. We test $H_0 : \mu = 0$ against $H_a : \mu < 0$.

(a) A test rejects when $X + Y < c$. Choose c so that the Type I error probability is 5%.

(b) A different test rejects if and only if both $X < -1$ and $Y < 0$. Find its power when $\mu = -1$.